

Week 3 Advanced Cybersecurity Practical Report

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Topic: Network Discovery, Traffic Analysis, Vulnerability Assessment, and Security Hardening

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➤ **Introduction :-**

This project presents a practical security assessment of an authorized internal lab network using Kali Linux, Nmap, and Wireshark in a VMware environment. The main purpose of the assessment was to understand how a cybersecurity analyst identifies devices on a network, discovers open services, analyzes network traffic, and improves security through hardening.

The assessment began with network discovery, where Nmap was used to find active devices and identify open ports and running services. The scan detected important services such as the DNS service (dnsmasq 2.45) and the Apache HTTP web server (Apache 2.4.68), which were later analyzed for possible security risks.

After identifying the services, Wireshark was used to capture and analyze network traffic. Normal activities such as web browsing, DNS lookups, and ping requests were performed to observe how devices communicate. Different protocols, including DNS, TCP, UDP, HTTP, ICMP, ARP, and TLS, were examined to understand packet flow and network behaviour.

The project also included a vulnerability assessment, where the scan results were used to identify security findings such as exposed services and insecure configurations. Finally, security hardening measures, including firewall configuration, service restriction, and system updates, were applied to reduce the attack surface and improve the overall security of the lab.

➤ **Importance of the Project: -**

- Helps understand how security analysts discover devices and services on a network.
- Develops practical skills in Nmap scanning and Wireshark traffic analysis.
- Improves understanding of common network protocols and communication patterns.
- Demonstrates how vulnerabilities can be identified using real scan results.
- Shows how security hardening reduces risks and strengthens network security.

Project Objectives: -

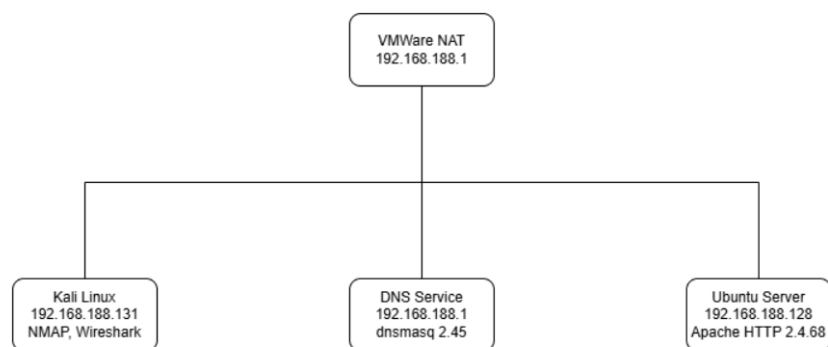
- Discover live devices connected to the lab network using Nmap.
- Identify open ports and running services on the detected devices.
- Capture and analyze network traffic using Wireshark.
- Study important network protocols such as DNS, TCP, UDP, HTTP, ICMP, and ARP.
- Understand how Nmap scans appear at the packet level in Wireshark.
- Identify security findings through a vulnerability assessment.
- Classify risks based on their potential impact on the network.
- Apply security hardening techniques, such as firewall rules and service restrictions.
- Compare the network's security before and after hardening.
- Learn the practical process followed by a security analyst during a basic network security assessment.

🚧 Lab Environment :-

- Lab Configuration

Component	Details
Hypervisor	VMware
Scanner	Kali Linux
Target	Ubuntu Server
Network	192.168.188.0/24
Kali IP	192.168.188.131
Ubuntu IP	192.168.188.128
DNS	192.168.188.2

🚧 Network Architecture:-



🚧 Tools Used

The following tools were used to perform network discovery, traffic analysis, vulnerability assessment, and security hardening in the authorized lab environment.

Tool	Purpose
Kali Linux	Used as the main security testing machine. It provides built-in cybersecurity tools for scanning and network analysis.
Nmap	Used to discover live hosts, identify open ports, detect running services, and perform security scans on authorized systems.
Wireshark	Used to capture and analyze network packets. It helped examine DNS, TCP, UDP, HTTP, ICMP, and ARP traffic.
VMware	Used to create a safe virtual lab with Kali Linux and Ubuntu machines for practical testing.
UFW Firewall	Used to apply firewall rules and restrict unnecessary network access during the security hardening task.
Apache HTTP Server	Used as the web server running on the Ubuntu machine for testing HTTP service and port scanning.
dnsmasq	Used as the DNS service in the lab to resolve domain names and observe DNS traffic during Wireshark analysis.

Important Commands Used

These commands were used during the practical tasks.

Command	Purpose
ip addr	Display the IP address and network interface details.
ip route	Show the default gateway and routing information.
nmap --version	Check the installed Nmap version.
nmap -sn 192.168.188.0/24	Discover live hosts on the network.
sudo nmap -sV 192.168.188.128	Detect service versions on the target machine.
sudo nmap -O 192.168.188.128	Detect the operating system of the target.
sudo nmap -sS 192.168.188.128	Perform a TCP SYN (half-open) scan.
sudo nmap -sU --top-ports 20 192.168.188.128	Scan the top 20 common UDP ports.
sudo nmap --script vuln 192.168.188.128	Run Nmap vulnerability detection scripts.
nslookup example.com	Generate DNS query traffic.
ping 192.168.188.128	Test network connectivity using ICMP.
sudo ufw enable	Enable the UFW firewall.
sudo ufw status	View the current firewall rules.
sudo apt update && sudo apt upgrade -y	Update and upgrade installed packages.

Nmap Methodology & Results :-

Methodology

The network assessment was performed using Nmap on the authorized lab network (192.168.188.0/24). The following steps were followed during the practical:

1. Checked the network configuration using `ip addr` to identify the IP address and subnet.
2. Verified that Nmap was installed using `nmap --version`.
3. Performed Host Discovery (`nmap -sn`) to find all active devices.
4. Performed Service Detection (`nmap -sV`) to identify running services and their versions.
5. Conducted a TCP SYN Scan (`nmap -sS`) to check which ports were open or closed.
6. Used Nmap Vulnerability Scripts (`nmap --script vuln`) to identify possible security issues.

These steps helped identify the devices, services, and potential security risks in the lab network.

Live Hosts Discovered

The host discovery scan found 5 live devices on the 192.168.188.0/24 network.

IP Address	Device/Result	Observation
192.168.188.1	Gateway	VMware NAT gateway device.
192.168.188.2	DNS Server	Running the DNS service.
192.168.188.128	Web Server	Hosting an Apache web server.
192.168.188.131	Kali Linux	Scanning machine used for testing.
192.168.188.254	Filtered Host	No open TCP ports were detected.

Observation: The scan confirmed that all these devices were active and reachable within the authorized lab network.

- **Open Services Identified**

The service detection scan identified the following important open services.

IP Address	Port	Service	Observation
192.168.188.2	53	DNS (dnsmasq 2.45)	Used for domain name resolution.
192.168.188.128	80	HTTP (Apache 2.4.68)	Web server available on the network.

Service Analysis: -

Port 53 – DNS

- The DNS service was running on 192.168.188.2.
- It converts domain names into IP addresses.
- This service is required for normal network communication.
- It should only be accessible to trusted devices.

Port 80 – HTTP

- The Apache web server was running on 192.168.188.128.
- It provides web pages and applications to users.
- Since it uses HTTP instead of HTTPS, the traffic is not encrypted.
- Access should be limited if the service is only for internal use.

Advanced Nmap Analysis

The advanced Nmap scan was performed to collect detailed information about the devices, operating systems, services, and possible security issues in the authorized lab network. Different Nmap options were used to identify service versions, detect operating systems, check UDP services, and run vulnerability scripts.

1. Version Detection (nmap -sV)

The -sV option was used to identify the software running on open ports.

Results

- 192.168.188.2 – dnsmasq 2.45 running on Port 53.
- 192.168.188.128 – Apache httpd 2.4.68 (Debian) running on Port 80.

Observation

- The DNS server is used for domain name resolution in the network.
- The Apache server is hosting a web service.
- Knowing the software version helps security analysts check for known vulnerabilities.

```
(kali@kali)-[~]
$ sudo nmap -sV 192.168.188.131/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 13:01 -0400
Nmap scan report for 192.168.188.1
Host is up (0.00047s latency).
All 1000 scanned ports on 192.168.188.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.188.2
Host is up (0.00021s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.45
MAC Address: 00:50:56:FD:96:40 (VMware)

Nmap scan report for 192.168.188.131
Host is up (0.00054s latency).
All 1000 scanned ports on 192.168.188.131 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:A3:15:47 (VMware)

Nmap scan report for 192.168.188.254
Host is up (0.00055s latency).
All 1000 scanned ports on 192.168.188.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E6:01:FA (VMware)

Nmap scan report for 192.168.188.128
Host is up (0.00016s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.68 ((Debian))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 256 IP addresses (5 hosts up) scanned in 29.40 seconds
```

2. Operating System Detection (nmap -O)

The -O option was used to identify the operating system of the target devices.

Results

- The web server was identified as a Linux-based Ubuntu/Debian system.
- The network gateway appeared to be a VMware NAT device.

Observation

- OS detection helps identify the type of device connected to the network.
- It also helps security analysts understand which security updates and configurations may be required.

```
(kali@kali)-[~]
└─$ sudo nmap -O 192.168.188.131/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 13:02 -0400
Nmap scan report for 192.168.188.1
Host is up (0.0049s latency).
All 1000 scanned ports on 192.168.188.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:c8:00:08 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.188.2
Host is up (0.0019s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 00:50:56:FD:96:40 (VMware)
Device type: specialized|general purpose
Running (JUST GUESSING): VMware Player (89%), Microsoft Windows XP|7|2012 (86%)
OS CPE: cpe:/a:vmware:player cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_s
erver_2012
Aggressive OS guesses: VMware Player virtual NAT device (89%), Microsoft Windows XP SP3 or Windows 7 or Windows Ser
ver 2012 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

Nmap scan report for 192.168.188.131
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.188.131 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:8C:29:A3:15:47 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.188.254
Host is up (0.00042s latency).
All 1000 scanned ports on 192.168.188.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E6:01:FA (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.188.128
Host is up (0.00036s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (96%), Linux 5.0 - 6.2 (96%), Linux 6.8 (96%), Linux 5.9 (95%), HP P2000 G3 NAS
```

3. UDP Scan (nmap -sU --top-ports 20)

The UDP scan checked the most common UDP ports used by network services.

Important UDP Services Found

Port	Service	Purpose
53	DNS	Domain name resolution
67	DHCP	Assigns IP addresses
123	NTP	Time synchronization
69	TFTP	File transfer service

Observation

- Most UDP ports were shown as open|filtered.
- This means Nmap could not fully confirm whether the service was open because UDP often does not send replies.
- These services should only be available if they are required by the network.

```

(kali@kali)-[~]
└─$ sudo nmap -sU --top-ports 20 192.168.188.131/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 13:06 -0400
Nmap scan report for 192.168.188.1
Host is up (0.0025s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
67/udp    open|filtered dhcp
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open|filtered netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   open|filtered netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
500/udp   open|filtered isakmp
514/udp   open|filtered syslog
520/udp   open|filtered route
631/udp   open|filtered ipp
1434/udp  open|filtered ms-sql-m
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
49152/udp open|filtered unknown
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.188.2
Host is up (0.0017s latency).

PORT      STATE      SERVICE
53/udp    open       domain
67/udp    open|filtered dhcp
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open|filtered netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   open|filtered netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
500/udp   open|filtered isakmp
514/udp   open|filtered syslog
520/udp   open|filtered route
631/udp   open|filtered ipp

```

4. NSE Vulnerability Scan (nmap --script vuln)

The Nmap Scripting Engine (NSE) was used to check for common vulnerabilities on the target system.

Result

- The scan found the /server-status page on the Apache web server.

Observation

- The server-status page can display information about the web server, such as active connections and server details.
- This information may help an attacker gather details about the system.

```

(kali@kali)-[~]
$ sudo nmap --script vuln 192.168.188.131/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 13:07 -0400
Nmap scan report for 192.168.188.1
Host is up (0.0092s latency).
All 1000 scanned ports on 192.168.188.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.188.2
Host is up (0.00016s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 00:50:56:FD:96:40 (VMware)

Nmap scan report for 192.168.188.131
Host is up (0.00091s latency).
All 1000 scanned ports on 192.168.188.131 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:A3:15:47 (VMware)

Nmap scan report for 192.168.188.254
Host is up (0.00036s latency).
All 1000 scanned ports on 192.168.188.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:E6:01:FA (VMware)

Nmap scan report for 192.168.188.128
Host is up (0.0019s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-enum:
|_ /server-status/: Potentially interesting folder

Nmap done: 256 IP addresses (5 hosts up) scanned in 61.87 seconds

```

Wireshark Methodology: -

Traffic Generation

During the practical, network traffic was generated in the authorized lab to capture different types of packets in Wireshark. The following activities were performed:

- Opened websites such as example.com and other web pages.
- Performed DNS lookups using nslookup and dig.
- Used the ping command to generate ICMP traffic.
- Accessed local services running in the lab, such as the Apache web server.
- Allowed the capture to run for about 5–10 minutes to collect enough network traffic.

These activities helped generate real network packets for analysis.

Wireshark Filters Used

Display filters were used to quickly find specific types of packets.

Filter	Purpose
dns	Shows only DNS query and response packets.

tcp	Displays all TCP communication.
udp	Shows UDP traffic such as DNS and NTP.
icmp	Displays Ping (Echo Request and Reply) packets.
arp	Shows ARP request and reply packets.
http	Displays HTTP requests and responses.

➤ Importance of Wireshark

- Captures live network traffic for analysis.
- Helps identify communication between devices.
- Makes it easier to investigate DNS, TCP, and HTTP traffic.
- Detects unusual network behavior during security investigations.
- Supports troubleshooting and network monitoring in cybersecurity.
- Provides packet-level evidence for vulnerability assessment and incident analysis.

Packet Analysis: -

During the Wireshark capture, different types of network packets were observed. These packets show how devices communicate with each other on the network. The analysis below explains the important events in simple words.

Important Packet Events

Time	Source IP	Destination IP	Protocol	Observation
7.2948	192.168.188.128	192.168.188.2	DNS	A DNS query was sent to find the IP address of a domain.
7.4921	89.111.28.78	192.168.188.128	TCP	A SYN/ACK packet confirmed that the connection request was accepted.
147.7347	192.168.188.128	8.8.8.8	ICMP	A ping request was sent to test network connectivity.
169.7041	Broadcast	LAN	ARP	ARP was used to find the MAC address of a device on the local network.
9.0317	10.5.28.229	50.19.115.217	HTTP	An HTTP request was sent to access a webpage.

Event Explanation

1. DNS Query (Time: 7.2948)

- The device 192.168.188.128 sent a DNS request to 192.168.188.2.
- The DNS server translated the domain name into an IP address.
- This is a normal step before opening a website.

Security Note: Normal DNS activity helps devices find websites, but unusual DNS requests should be investigated.

2. TCP SYN/ACK Response (Time: 7.4921)

- The server 89.111.28.78 replied with a SYN/ACK packet.
- This shows that the TCP connection request was accepted.
- It is part of the TCP three-way handshake.

Security Note: A SYN/ACK response usually means the destination service is available and accepting connections.

3. ICMP Ping Request (Time: 147.7347)

- The system sent a ping request to 8.8.8.8.
- This was used to check whether the network connection was working.
- The successful reply confirmed internet connectivity.

Security Note: Ping traffic is normal, but excessive ICMP traffic may require investigation.

4. ARP Address Resolution (Time: 169.7041)

- An ARP request was broadcast on the local network.
- It asked for the MAC address of another device.
- This allowed communication inside the LAN.

Security Note: ARP is a normal network process, but unexpected ARP replies could indicate ARP spoofing.

5. HTTP Web Request (Time: 9.0317)

- A browser sent an HTTP request to a web server.
- The server responded successfully.
- This shows normal web browsing activity.

Security Note: HTTP traffic is not encrypted, so HTTPS should be used for better security.

 **Nmap + Wireshark Investigation: -**

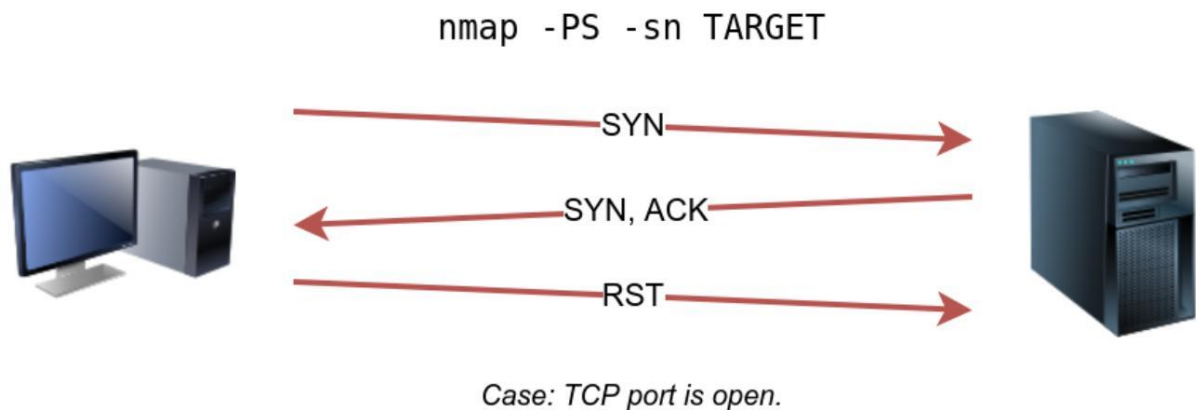
Overview: -

During this task, Nmap was used to perform a TCP SYN scan, while Wireshark captured the packets in real time. This helped verify how Nmap detects open, closed, and filtered ports by analyzing TCP packets.

SYN Scan Analysis: -

Wireshark confirmed that Nmap used the TCP three-way handshake to identify port status.

How the SYN Scan Worked: -



1. Nmap sent a SYN packet to the target.
2. Open ports replied with SYN/ACK.
3. Nmap immediately sent an RST packet instead of completing the connection.
4. This is called a Half-Open (SYN) Scan, which is faster than a full TCP connection.

Nmap vs Wireshark Comparison :-

Nmap Result	Wireshark Evidence	Meaning
Port 53 Open	SYN → SYN/ACK	DNS service is reachable.
Port 80 Open	SYN → SYN/ACK	Apache web server is reachable.
Port 199 Closed	SYN → RST	The port is closed and rejected the connection.
Port 587 Closed	SYN → RST	SMTP Submission port is not accepting connections.
Filtered Host	No response	Traffic was likely blocked by a firewall or filtering device.

Key Observations: -

- The scanning host sent SYN packets to multiple ports such as 53, 80, 22, 199, and 587.
- Port 53 (DNS) responded with SYN/ACK, confirming that the DNS service was running.

- Port 80 (Apache HTTP) also responded with SYN/ACK, showing that the web server was active.
- Closed ports returned RST packets.
- Some hosts did not respond, indicating possible firewall filtering.

Vulnerability Assessment & Risk Classification: -

Medium Risk Findings: -

1. HTTP Port 80 Exposed – Medium Risk

Evidence: Nmap detected Apache HTTP 2.4.68 running on Port 80.

Why Medium?

- The web server can be accessed by devices on the network.
- HTTP does not encrypt data.
- Sensitive information could be exposed if HTTP is used.

Potential Impact

- Attackers may access the web service.
- Data can be intercepted during communication.

Recommendation

- Enable HTTPS.
- Restrict access using firewall rules.
- Keep Apache updated.

2. /server-status Exposed – Medium Risk

Evidence: Nmap vulnerability scan found the /server-status page.

Why Medium?

- The page may display server information.
- Attackers could use this information to learn about the server.

Potential Impact

- Information disclosure.
- Easier reconnaissance for attackers.

Recommendation

- Restrict the page to localhost.
- Disable it if it is not needed.

3. DNS Port 53 Accessible – Medium Risk

Evidence: Nmap detected dnsmasq 2.45 on Port 53.

Why Medium?

- DNS is an important service but should only be available to trusted devices.
- If it is not restricted, it may expose network information.

Potential Impact

- Information gathering.
- Possible misuse of DNS requests.

Recommendation

- Allow DNS access only from trusted devices.
- Keep dnsmasq updated.
- Monitor DNS activity.

4. HTTP Traffic Observed – Medium Risk

Evidence: Wireshark captured HTTP GET requests and responses.

Why Medium?

- HTTP sends data without encryption.
- Anyone monitoring the network could read the traffic.

Potential Impact

- User information may be exposed.
- Login details could be intercepted if HTTP is used.

Recommendation

- Use HTTPS instead of HTTP.
- Redirect users to secure connections.

Low Risk Finding

5. Repeated SYN Packets – Low Risk

Evidence: Wireshark showed multiple TCP SYN packets during the Nmap scan.

Why Low?

- The activity came from an authorized Nmap scan.
- Similar behavior should still be monitored in a real network.

Potential Impact

- Could indicate scanning activity if performed by an unknown device.

Recommendation

- Monitor repeated SYN packets.
- Configure firewall or SIEM alerts for unusual scanning activity.

Security Hardening: -

Overview: -

After completing the Nmap and Wireshark analysis, I improved the security of my authorized lab by fixing the issues found during the assessment. The goal was to reduce unnecessary network exposure, protect sensitive services, and verify the changes using Nmap.

Security Hardening Steps

1. Restrict Port 80 using UFW

- Port 80 was open on the Ubuntu server.
- I enabled the UFW (Uncomplicated Firewall).
- Access to Port 80 was limited to the trusted local network (192.168.188.0/24).
- This reduced unnecessary access from other devices.

2. Secure the /server-status Page

- The Nmap vulnerability scan found the /server-status page.
- I changed the Apache configuration to allow only local access.
- This prevents unauthorized users from viewing server information.

3. Update System Packages

- I updated the operating system packages.
- Apache and other installed packages were upgraded.
- Regular updates help fix security issues and improve system stability.

Commands Used

- `sudo ufw enable`
- `sudo ufw allow from 192.168.188.0/24 to any port 80`
- `sudo systemctl restart apache2`
- `sudo apt update`
- `sudo apt upgrade -y`

Why These Changes Were Important

- Reduced unnecessary network exposure.
- Protected sensitive server information.
- Improved access control using firewall rules.
- Kept the system updated with security patches.

- Made the lab more secure without affecting normal services.

Before vs After Table

Security Issue	Before	Action Taken	After	Improvement
Port 80 (HTTP)	Port 80 was open and accessible on the network.	Restricted access using UFW firewall rules.	Access became limited to the trusted local network.	Reduced unnecessary network exposure.
/server-status Page	The page was visible during the Nmap vulnerability scan.	Changed Apache configuration to Require local and restarted the service.	The page became accessible only from the local system.	Prevented server information from being exposed.
System Packages	Existing packages needed maintenance.	Ran sudo apt update and sudo apt upgrade -y.	The system was updated and services continued working normally.	Improved security by installing the latest updates.

- **Attack Surface Reduction: -**

The hardening process reduced the attack surface of the lab network in several ways:

- Restricted web server access to trusted users.
- Protected the /server-status page from unauthorized access.
- Updated system packages to improve security and stability.
- Reduced unnecessary network exposure.
- Made the lab environment more secure while keeping required services operational.

- **Security Recommendations**

The following recommendations can help improve the security of the lab network.

1. Enable HTTPS for Web Services: Replace HTTP with HTTPS so that data between the user and the server is encrypted and protected.
2. Restrict Unnecessary Ports: Allow only required ports through the firewall and close unused ports to reduce the attack surface.
3. Keep Software Updated: Regularly update the operating system, Apache server, and other software to fix security vulnerabilities.
4. Protect Sensitive Pages: Restrict pages such as /server-status so that only authorized users can access them.
5. Monitor Network Traffic: Use Wireshark or other monitoring tools to detect unusual traffic, repeated connection attempts, or unexpected network activity.

- **Challenges Faced**

During this practical, I faced a few challenges while performing the security assessment.

- Selecting the correct network interface in Wireshark for packet capture.
- Understanding TCP flags such as SYN, SYN/ACK, and RST.
- Identifying the difference between open, closed, and filtered ports in Nmap.
- Distinguishing normal network traffic from traffic that required further investigation.
- Configuring firewall rules correctly without affecting required services.

These challenges helped me improve my practical understanding of network security tools.

- **Conclusion**

This project provided practical experience in performing a basic security assessment of an internal network. I used Nmap to discover live hosts, identify open ports, detect running services, and perform advanced scanning. I then used Wireshark to capture and analyze network traffic, including DNS, TCP, UDP, HTTP, ICMP, and ARP packets.

Based on the collected evidence, I identified several security findings, performed a vulnerability assessment, and applied security hardening by restricting services and improving configurations. Finally, I compared the results before and after hardening to verify that the attack surface had been reduced.

Overall, this project improved my skills in network scanning, packet analysis, vulnerability assessment, and security hardening, giving me a better understanding of how security analysts investigate and secure a network.

- **References**

The following tools and official documentation were used during this practical.

- Nmap – Network Scanning Tool
- Wireshark – Network Packet Analyzer
- Kali Linux
- Ubuntu Server
- VMware Workstation
- Apache HTTP Server Official Documentation
- UFW (Uncomplicated Firewall) Official Documentation

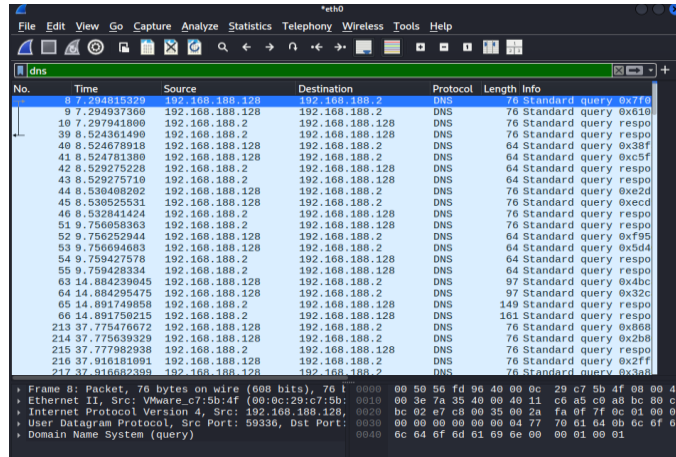
- **Screenshots & Evidence: -**

The appendix contains the screenshots collected during Tasks 7–13 as evidence of the practical work.

Nmap Evidence

Wireshark Evidence

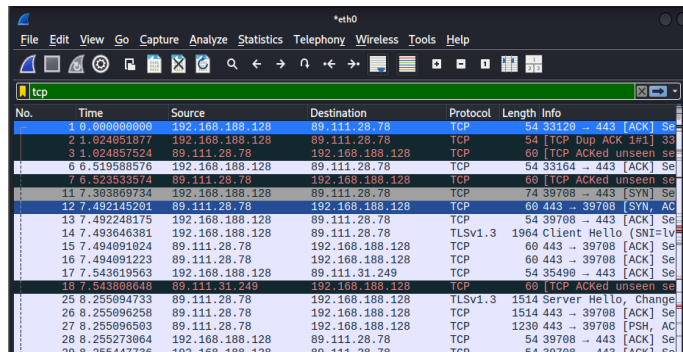
- DNS Query and Response



A screenshot of the Wireshark network protocol analyzer showing a capture of DNS traffic. The packet list pane on the left shows 217 packets, with the selected packet (No. 8) being a DNS Standard query. The packet details pane on the right shows the structure of the DNS query, including the transaction ID (0x7f0), flags (Standard query), and the question section (Standard query 0x7f0).

No.	Time	Source	Destination	Protocol	Length	Info
8	7.294815329	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
9	7.294937360	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
10	7.297941800	192.168.188.2	192.168.188.128	DNS	76	Standard query 0x7f0
39	8.524361490	192.168.188.2	192.168.188.128	DNS	76	Standard query 0x7f0
40	8.524678918	192.168.188.128	192.168.188.2	DNS	64	Standard query 0x7f0
41	8.524781300	192.168.188.128	192.168.188.2	DNS	64	Standard query 0x7f0
42	8.529275228	192.168.188.2	192.168.188.128	DNS	64	Standard query 0x7f0
43	8.529275710	192.168.188.2	192.168.188.128	DNS	64	Standard query 0x7f0
44	8.530480202	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
45	8.530525531	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
46	8.532841424	192.168.188.2	192.168.188.128	DNS	76	Standard query 0x7f0
51	9.750058363	192.168.188.2	192.168.188.128	DNS	76	Standard query 0x7f0
52	9.750252944	192.168.188.128	192.168.188.2	DNS	64	Standard query 0x7f0
53	9.750694683	192.168.188.128	192.168.188.2	DNS	64	Standard query 0x7f0
54	9.750427578	192.168.188.2	192.168.188.128	DNS	64	Standard query 0x7f0
55	9.750428334	192.168.188.2	192.168.188.128	DNS	64	Standard query 0x7f0
63	14.884239845	192.168.188.128	192.168.188.2	DNS	97	Standard query 0x7f0
64	14.884295475	192.168.188.128	192.168.188.2	DNS	97	Standard query 0x7f0
65	14.891749858	192.168.188.2	192.168.188.128	DNS	149	Standard query 0x7f0
66	14.891760215	192.168.188.2	192.168.188.128	DNS	161	Standard query 0x7f0
213	37.775476672	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
214	37.775639329	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
215	37.777982938	192.168.188.2	192.168.188.128	DNS	76	Standard query 0x7f0
216	37.916181091	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0
217	37.916682358	192.168.188.128	192.168.188.2	DNS	76	Standard query 0x7f0

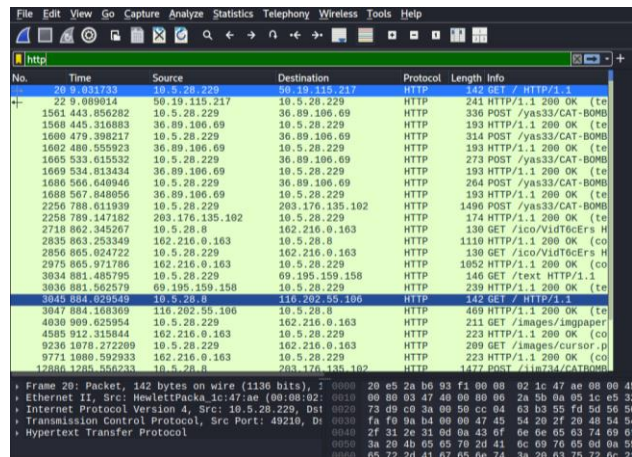
- TCP Handshake (SYN, SYN/ACK, RST)



A screenshot of the Wireshark network protocol analyzer showing a capture of TCP traffic. The packet list pane on the left shows 26 packets, with the selected packet (No. 1) being a TCP SYN. The packet details pane on the right shows the structure of the TCP SYN, including the source port (54312) and destination port (443).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.188.128	89.111.28.78	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
2	1.024051077	192.168.188.128	89.111.28.78	TCP	54	[TCP Dup ACK 1 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
3	1.024857524	89.111.28.78	192.168.188.128	TCP	60	[TCP ACKed unseen seq=1000000000, Win=0, Len=0]
6	6.519588576	192.168.188.128	89.111.28.78	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
7	6.523533574	89.111.28.78	192.168.188.128	TCP	60	[TCP ACKed unseen seq=1000000000, Win=0, Len=0]
11	7.3033069134	192.168.188.128	89.111.28.78	TCP	74	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
12	7.492145291	89.111.28.78	192.168.188.128	TCP	60	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
13	7.492248175	192.168.188.128	89.111.28.78	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
14	7.493646381	192.168.188.128	89.111.28.78	TLSv1.3	1964	Client Hello [SN=1, Ver=3, Ciphers=...
15	7.494091024	89.111.28.78	192.168.188.128	TCP	60	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
16	7.494091223	89.111.28.78	192.168.188.128	TCP	60	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
17	7.543619563	192.168.188.128	89.111.31.249	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
18	7.543888648	89.111.31.249	192.168.188.128	TCP	60	[TCP ACKed unseen seq=1000000000, Win=0, Len=0]
25	8.255094733	89.111.28.78	192.168.188.128	TLSv1.3	1514	Server Hello, Change...
26	8.255096258	89.111.28.78	192.168.188.128	TCP	1514	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
27	8.255096503	89.111.28.78	192.168.188.128	TCP	1230	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
28	8.255273964	192.168.188.128	89.111.28.78	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]
29	8.255477736	192.168.188.128	89.111.28.78	TCP	54	[TCP SYN 54312 → 443 [ACK] Seq=1000000000, Win=0, Len=0]

- HTTP Traffic



A screenshot of the Wireshark network protocol analyzer showing a capture of HTTP traffic. The packet list pane on the left shows 26 packets, with the selected packet (No. 1) being an HTTP GET. The packet details pane on the right shows the structure of the HTTP GET, including the source port (54312) and destination port (443).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.031733	10.5.28.229	10.5.28.229	HTTP	142	GET / HTTP/1.1
2	9.089014	10.5.28.229	10.5.28.229	HTTP	241	HTTP/1.1 200 OK (text/html)
3	1561.443.856282	10.5.28.229	10.5.28.229	HTTP	336	POST /yas33/CAT-BOMB
4	1568.445.310883	36.89.186.69	10.5.28.229	HTTP	193	HTTP/1.1 200 OK (text/html)
5	1600.479.398217	10.5.28.229	36.89.186.69	HTTP	314	POST /yas33/CAT-BOMB
6	1602.480.555923	36.89.186.69	10.5.28.229	HTTP	193	HTTP/1.1 200 OK (text/html)
7	1605.530.615532	10.5.28.229	36.89.186.69	HTTP	273	POST /yas33/CAT-BOMB
8	1609.534.813434	36.89.186.69	10.5.28.229	HTTP	193	HTTP/1.1 200 OK (text/html)
9	1680.560.640940	10.5.28.229	36.89.186.69	HTTP	264	POST /yas33/CAT-BOMB
10	1688.567.848056	36.89.186.69	10.5.28.229	HTTP	193	HTTP/1.1 200 OK (text/html)
11	2250.788.611939	10.5.28.229	263.176.135.102	HTTP	1490	POST /yas33/CAT-BOMB
12	2258.789.147182	263.176.135.102	10.5.28.229	HTTP	174	HTTP/1.1 200 OK (text/html)
13	2718.862.345267	10.5.28.8	162.216.0.163	HTTP	130	GET /ico/vidt6cErH
14	2835.863.253349	162.216.0.163	10.5.28.8	HTTP	1110	HTTP/1.1 200 OK (text/html)
15	2856.865.604722	10.5.28.229	162.216.0.163	HTTP	130	GET /ico/vidt6cErH
16	2975.865.971786	162.216.0.163	10.5.28.229	HTTP	1052	HTTP/1.1 200 OK (text/html)
17	3034.881.485795	10.5.28.229	69.195.159.158	HTTP	146	GET /text HTTP/1.1
18	3036.881.582579	69.195.159.158	10.5.28.229	HTTP	239	HTTP/1.1 200 OK (text/html)
19	3045.884.629549	10.5.28.8	110.202.55.106	HTTP	142	GET / HTTP/1.1
20	3047.884.168369	110.202.55.106	10.5.28.8	HTTP	469	HTTP/1.1 200 OK (text/html)
21	4030.909.625954	10.5.28.229	162.216.0.163	HTTP	211	GET /images/impaper
22	4985.912.315044	162.216.0.163	10.5.28.229	HTTP	223	HTTP/1.1 200 OK (text/html)
23	9236.1078.272209	10.5.28.229	162.216.0.163	HTTP	209	GET /images/cursor.p
24	9771.1080.592933	162.216.0.163	10.5.28.229	HTTP	223	HTTP/1.1 200 OK (text/html)
25	12906.4285.256233	10.5.28.8	263.176.135.102	HTTP	142	POST /lin23/CAT-BOMB

- ICMP Ping

Wireshark packet capture showing ICMP Echo (ping) requests and replies. The interface is set to *eth0. The filter is icmp. The packet list shows several requests and replies.

No.	Time	Source	Destination	Protocol	Length	Info
491	147.734784826	192.168.188.128	8.8.8.8	ICMP	98	Echo (ping) request id=
492	147.748924934	8.8.8.8	192.168.188.128	ICMP	98	Echo (ping) reply id=
493	148.733719199	192.168.188.128	8.8.8.8	ICMP	98	Echo (ping) request id=
494	148.748019492	8.8.8.8	192.168.188.128	ICMP	98	Echo (ping) reply id=
495	149.734116683	192.168.188.128	8.8.8.8	ICMP	98	Echo (ping) request id=
496	149.749505181	8.8.8.8	192.168.188.128	ICMP	98	Echo (ping) reply id=
497	150.734962405	192.168.188.128	8.8.8.8	ICMP	98	Echo (ping) request id=
498	150.751475602	8.8.8.8	192.168.188.128	ICMP	98	Echo (ping) reply id=

- ARP Requests and Replies

Wireshark packet capture showing ARP requests and replies. The interface is set to *eth0. The filter is arp. The packet list shows several requests and replies.

No.	Time	Source	Destination	Protocol	Length	Info
521	169.704177192	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
534	170.711863419	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
539	171.735958719	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
540	172.760323814	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
543	173.783870576	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
546	174.808109831	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.121?
549	189.693406850	VMware_fd:96:40	Broadcast	ARP	60	Who has 192.168.188.128?
550	189.693657965	VMware_fd:96:40	VMware_fd:96:40	ARP	42	192.168.188.128 is at 00
625	233.695441640	VMware_c7:5b:4f	Broadcast	ARP	42	Who has 192.168.188.2? T
626	233.696217888	VMware_fd:96:40	VMware_c7:5b:4f	ARP	60	192.168.188.2 is at 00:5